

Feature Extraction and Visualization of MI-EEG by LLE Algorithm

Ming-ai Li, Xinyong Luo, Meng Zhang, Jinfu Yang

*College of Electronic Information & Control Engineering, Beijing University of
Technology, Beijing 100124, China*

limingai@bjut.edu.cn, luoxinyong163@163.com, vzhangmeng726@163.com,
jfyang@bjut.edu.cn

Abstract - As a nonlinear time-varying and non-stationary signal, Motor Imagery Electroencephalography (MI-EEG) has attracted many researchers to use its time-frequency feature extraction by using Discrete Wavelet Transform (DWT) in brain computer interfaces (BCIs). Though a few people have devoted their efforts to exploring its nonlinear nature from the perspective of manifold learning, they hardly take into full account both time-frequency feature and nonlinear nature. To obtain features that can fully describe the information from a nonlinear nature and time-frequency perspective of MI-EEG, a novel feature extraction method is proposed based on the Locally Linear Embedding algorithm (LLE) and DWT. The multi-scale multi-resolution analysis is implemented for MI-EEG with DWT, and the valid time and frequency windows are determined in advance by a Wigner-Ville distribution. In view of the nonlinear structure in MI-EEG, LLE is applied to the approximation components to obtain the nonlinear features, and the statistics of the detail components are calculated to obtain the time-frequency features. After an organic combination of the two features, a Back-Propagation neural network optimized by a Genetic Algorithm was employed as a classifier to evaluate the effectiveness of the proposed feature extraction method. Compared with conventional DWT-based methods, the proposed method has a better effect on feature visualization with an obvious clustering distribution and improves the classification results and their stability. This paper successfully achieves manifold learning in signal processing of EEG.

Index Terms - feature extraction, locally linear embedding, motor imagery electroencephalography, visualization, BP neural network.

I. INTRODUCTION

In response to an external stimulus or active thinking activity, nerve cells in the brain cortex can produce Motor Imagery Electroencephalography (MI-EEG) signals with the characteristics of specificity and rhythmicity, and they not only contain a large number of physiological or disease information but also have a close correlation with the state of consciousness. As a result, much attention has been paid to the application of MI-EEG in brain cognition; meanwhile, the correct interpretation and accurate extraction of MI-EEG features is the key to its successful applications [1-2].

MI-EEG has the characteristics of individual differences, nonlinear, non-stationary and time-varying sensitive. Wavelet Transform (WT) is able to take advantage of scale and shift operations to perform multi-scale decomposition and time-frequency domain localization, effectively obtaining the time-frequency information of signals. Thus, the analysis of EEG

signals can benefit from WT. Despite this, recent studies do not support the effective use of wavelet features for the discrimination of EEG signals because of the redundant and irrelevant information contained in wavelet coefficients [3]. To this day, these characteristics remain a problem for the extraction of useful features from EEG signals for classification. Many approaches to the problem were proposed in the last decade. Xu proposed a novel method of extracting EEG features based on discrete wavelet transform (DWT) and autoregressive (AR), in which the combination features of wavelet coefficient statistics and the sixth-order AR coefficients were used as input vectors for the classifier [4]. To solve the problem of a lack of spatial information and large amounts of features, a method based on the time-frequency-space domain was proposed using independent component analysis (ICA) and wavelet transform [5]. A method using only wavelet coefficients from the sub-band bound with specific frequency features of MI-EEG was proposed by Monica[6]. R. K. Chaurasiya performed a linear dimensionality reduction of wavelet coefficients by principal component analysis (PCA) and then used the coordinates embedded in the low dimension space as features and obtained better classification results[7]. Obviously, wavelet transform, as a type of classical time-frequency analysis method, played an important role in the feature extraction of the EEG signal.

However, the brain is a highly complex biological system in terms of structure and function, namely, the brain is a typical nonlinear system [8]. MI-EEG signals not only contain abundant information and rhythmic characteristics of motor imagery consciousness but also possess an obvious nonlinear structure. Taking the reason that traditional time-frequency feature extraction methods were based on linear system theory into consideration, we will inevitably find a loss of information in the original signal, and follow-up questions about the nonlinear structural characteristics cannot be excavated from high-dimensional EEG datasets [9]. Manifold learning (ML) can recover the structure of lower dimensional manifolds from high dimensional data and can help us obtain the corresponding nonlinear embedded coordinates that are regarded as a meaningful representation of reduced data. During the dimension reduction process, ML can best preserve the local neighborhood of each object or, in other words, the potential of the manifold structure. At present, Locally Linear Embedding (LLE), as one of typical ML algorithms, has got a certain application in feature mapping of EEG signal, for the reason that LLE as a non-iterative algorithm has low

computing complexity and local optimal analytical solutions. Felix Lee[10] calculated MI-EEG signal adaptive autoregression model parameters based on kalman filter, and then by using LLE performed dimension reduction to get EEG features, and the classification accuracy was between 61.9% and 74.2% with a linear discriminant analysis classifier; Yaozhang Pan [11] extracted the features of EEG by weighted LLE for detecting epilepsy, and the classification rate was between 72.79% and 90.66% using SVM classifier; Mirsadeghi M [12] calculated the power, covariance, and a variety of entropy of EEG to obtain the primitive features, and LLE was applied to reduce the feature dimension. Then the quadratic discriminant analysis (QDA) was adopted to estimate the depth of anesthesia, the average classification accuracy was 88.4%. From the studies, we can conclude that LLE is applicable for feature extraction and feature dimension reduction of EEG. However, there exist the following problems need to be improved: (1) LLE algorithm is sensitive to data noise, so if it is directly used in feature extraction and feature dimension reduction of EEG signal, the low dimensional embedding structure may be easily destroyed, which affects the quality of EEG features; (2) Because EEG signal has significant time-frequency distribution and nonlinear characteristics, using only ML algorithm is difficult to get its essential characteristics comprehensively, which sometimes even will cause feature information redundancy and mismatching problem.

In this paper, to achieve a comprehensive and accurate extraction of the essential features of MI-EEG signals, a feature extraction method was proposed based on a Locally Linear Embedding algorithm (LLE), a typical manifold learning algorithm, and Discrete Wavelet Transform. We used DWT to obtain the time-frequency feature and then applied LLE to the approximation components to obtain nonlinear features. Finally, the time-frequency features and nonlinear features were fused in serial, and then, a BP neural network optimized by a genetic algorithm was employed as a classifier and many experiments were conducted on a publicly available dataset with 10-fold cross validation. Compared with the traditional DWT-based feature extraction methods, the proposed method enabled the feature vector to contain more comprehensive and complete MI-EEG information because of the addition of nonlinear features. Thus, better classification results have been achieved and demonstrate the validity of this method.

II. ALGORITHM

A. Discrete Wavelet Transform

The time-frequency domain localization and multi-resolution as characteristics of wavelet transform make it possible obtain transient information in different scales under different frequencies that are suitable for the analysis of non-stationary signals, such as EEG. This paper uses discrete wavelet transform to analyze the Event-Related Desynchronization (ERD) and Event-Related Synchronization (ERS) of MI-EEG for right and left hand movements.

Based on the Mallat algorithm, for the L-level wavelet decomposition of the signal, the given signal $f(t)$ can be represented as:

$$f(t) = A_L + \sum_{j=1}^L D_j \quad (1)$$

where L denotes the number of the decomposition level, A_L denotes the approximation coefficients and D_j denotes the detail coefficients at scale $j, j=1, 2, \dots, L$. Thus, the signal $f(t)$ can be divided into several sub-bands. Assuming that the sampling frequency of the signal is f_s , then for the coefficients $A_L, D_L, D_{L-1} \dots D_1$, the corresponding frequency band ranges are $\left[0, \frac{f_s}{2^{L+1}}\right], \left[\frac{f_s}{2^{L+1}}, \frac{f_s}{2^L}\right], \left[\frac{f_s}{2^L}, \frac{f_s}{2^{L-1}}\right], \dots, \left[\frac{f_s}{2^2}, \frac{f_s}{2}\right]$, respectively. In wavelet decomposition for MI-EEG, the number of decomposition layers is decided by the distribution of useful components in the signal in the frequency domain and the sampling frequency.

B. Locally Linear Embedding Algorithm

The locally linear embedding algorithm is a type of typical manifold learning algorithm. The main idea of LLE is to solve globally nonlinear problems using locally linear fitting, which is based on the assumption that data lying on a nonlinear manifold can be viewed as linear in local areas [13]. By computing low-dimensional, neighbor-hood-preserving embeddings of high-dimensional as inputs, LLE maps its inputs into a single global coordinate system of lower dimensionality, and its optimizations do not involve local minima. By exploiting the local symmetries of linear reconstructions, LLE is able to learn the global structure of nonlinear manifolds. [14] The locally linear fitting is characterized by linear coefficients that reconstruct each data point from its neighbors and the linear coefficients, namely, the weight matrix; record the neighborhood information; and establish a bridge between the high-dimensional data space and the low-dimensional latent space.

The Locally Linear Embedding algorithm consists of three steps:

- 1) Select neighbors: as an input, LLE takes a set of n D-dimension vectors assembled in a matrix $X = [x_1, x_2, \dots, x_n] \in R^{D \times n}$. According to the similarities between data points measured by Euclidean distances, for each sample $x_i, i = 1, \dots, n$, its k nearest neighbors are sought, and then, their indices are stored in an $n \times k$ matrix J .
- 2) Reconstruct with linear weights: to find the reconstruction weight matrix $W \in R^{n \times n}$, where w_{ij} represents the contribution of the data point x_j to the data point x_i , the following cost function should be minimized.

$$e(W) = \sum_{i=1}^n \left\| x_i - \sum_{j=1}^n w_{ij} x_j \right\|^2 \quad (2)$$

where the optimization should be subject to constraints

$$\sum_{j=1}^n w_{ij} = 1 \quad (3)$$

and $w_{ij} = 0$, if x_i and x_j are not neighbors. If not, w_{ij} is decided by the optimization of reconstruction errors in the way of solving a least-squares problem (2).

- 3) Map to embedded coordinates: the weight matrix W is kept fixed and the low-dimensional embedding $Y = [y_1, y_2, \dots, y_n] \in R^{d \times n}$, where d represents embedded dimension, can be obtained by minimizing the embedding cost function

$$\Phi(Y) = \sum_{i=1}^n \left\| y_i - \sum_{j=1}^n w_{ij} y_j \right\|^2 \quad (4)$$

Under the constraints below:

$$\sum_{i=1}^n y_i = 0 \quad (5)$$

$$\frac{1}{N} \sum_{i=1}^n y_i y_i^T = I \quad (6)$$

Which provide a unique solution. A new matrix M of size $n \times n$ is constructed based on the matrix W : $M = (I - W)^T(I - W)$, and then, we can obtain the low-dimensional embedding Y by carrying out an eigen-decomposition of the new sparse matrix M [14]. Because the first eigenvector, whose eigenvalue is close to zero, is excluded, Y can be represented as below

$$Y = [v_2, \dots, v_{d+1}]^T \quad (7)$$

where $v_2, \dots, v_{d+1}$ denotes the 2^{nd} to $(d+1)^{\text{st}}$ smallest eigenvalues of matrix M .

III. DESCRIPTION OF FEATURE EXTRACTION

A. Experimental Data Description

The experimental dataset was from BCI Competition 2003 provided by BCI Lab, Graz University of Technology (<http://www.bbc.de/competition/iii/>). The dataset was composed of 280 trials, of which 140 were for training and 140 were used for testing images of left/right hand movements. Each trial lasted for 9 seconds, and the sequence diagram of the experiment is shown schematically in Fig. 1.

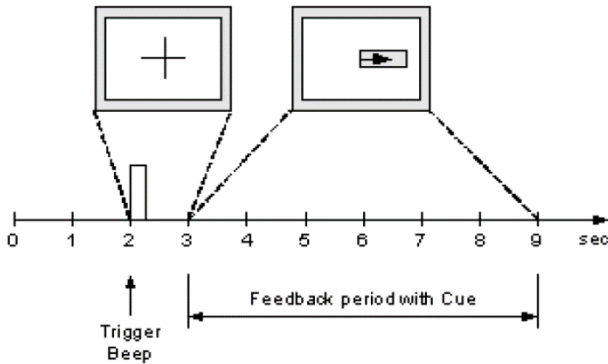


Fig. 1: Timing Scheme of the Experiment

The first 2 seconds were quiet, during which the subjects kept calm and a short beep indicated the start of the trial at the end of second 2, with a fixation cross '+' displayed on the

screen simultaneously. At the third second, the visual cue (left-right arrow) was displayed as the direction of motor imagery. The data were sampled at 128 Hz. Three channels were measured over a C3, CZ and C4 conductor, using Ag/AgCl as an electrode, and the placement of the electrode is shown in Fig. 2.

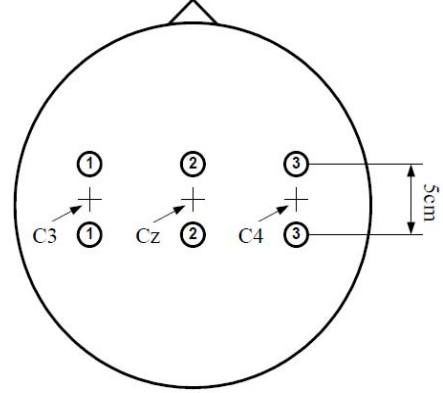


Fig. 2: Electrode Placement

B. Time-Frequency Feature

Considering the characteristics of the rhythmic distribution in the time-frequency domain of MI-EEG, db5 wavelet function in Class Daubechies was used to make a 3-level wavelet decomposition of MI-EEG, which provides the information at different resolutions of the samples at different frequency bands [15]. After the decomposition, A_3 denotes its approximation coefficients and D_1 - D_3 denote its detail components. Because the wavelet coefficients express the energy distribution of the signal in the time-frequency domain [4] and rhythm Alpha (8~13Hz) is close to D_3 , whose frequency range is 8~16 Hz, and rhythm Beta (14 ~ 30 Hz) is similar to D_2 , whose frequency range is 16~30 Hz, the two frequency ranges are the most obvious frequency bands for the ERD and ERS phenomenon [16]. Considering the frequency of our interest, the detail components D_2 and D_3 are chosen for feature extraction.

To depict the feature in the view of signal energy, the statistics of the detail components are calculated as the time-frequency feature and the statistical features, such as the average, mean value of energy and mean value of the standard deviation of the detail components. Assume that $x_{j,k}^i \in R^{1 \times n}$ represents the detail components D_j ($i = 3, 4$ $j = 2, 3$) from channel C_i of MI-EEG and $i = 1, 2, \dots, n$, where n represents the number of detail components. Thus, the time-frequency feature can be obtained as shown below, and the average is computed as follows:

$$\bar{\mu}_j^i = \frac{1}{n} \sum_{k=1}^n x_{j,k}^i \quad (8)$$

The mean value of energy is determined by Eq. (9).

$$\bar{E}_j^i = \frac{1}{n} \sum_{k=1}^n (x_{j,k}^i)^2 \quad (9)$$

And the mean value of the standard deviation is calculated by

$$S_j^i = \sqrt{\frac{1}{n} \sum_{k=1}^n (x_{j,k}^i - \bar{\mu}_j^i)^2} \quad (10)$$

Considering the performance phenomenon ERD and ERS in channels C3 and C4, to make the time-frequency feature show a sharp distinction, the time-frequency feature $F_1 \in R^{6 \times 1}$ is defined as

$$F_1 = \left[\frac{\bar{\mu}_2^3 - \bar{\mu}_2^4}{\|\bar{\mu}_2^3 - \bar{\mu}_2^4\|}, \frac{\bar{\mu}_3^3 - \bar{\mu}_3^4}{\|\bar{\mu}_3^3 - \bar{\mu}_3^4\|}, \frac{\bar{E}_2^3 - \bar{E}_2^4}{\|\bar{E}_2^3 - \bar{E}_2^4\|}, \frac{\bar{E}_3^3 - \bar{E}_3^4}{\|\bar{E}_3^3 - \bar{E}_3^4\|}, \frac{S_2^3 - S_2^4}{\|S_2^3 - S_2^4\|}, \frac{S_3^3 - S_3^4}{\|S_3^3 - S_3^4\|} \right]^T \quad (11)$$

where $\|\cdot\|$ represents the second order norm.

C. Nonlinear Feature

To fully acquire the accurate nonlinear information of the distribution of MI-EEG, LLE is used to extract features, specifically, features that can loyally represent the nonlinear information in a high-dimensional observation space and simultaneously reduce nonlinear dimensionality and visualization in the view of manifold learning.

(1) The Nonlinear Feature Based on Original MI-EEG

First, the MI-EEG from channels C3 and C4 during the 3.5~7 s period, in which the ERD and ERS phenomena are remarkable, are concatenated together; then, a nonlinear dimensionality reduction is performed based on the LLE and the low-dimensional embedding is Y_2 . Thus, the maximum differentiation of three coordinates was chosen as the nonlinear feature F_2 and is defined as

$$F_2 = Y_2^* \in R^{3 \times 1} \quad (12)$$

The 3-D visualization of F_2 is shown in Fig. 3.

(2) The Nonlinear Feature Based on Wavelet Coefficients

First, we performed a 3-level wavelet decomposition of MI-EEG; then, because the approximation components reflect the time-frequency feature distribution of a low-frequency signal in the original MI-EE and the high-frequency details contain more noise, in the process of nonlinear dimensionality reduction from the embedded high-dimensional observation space to the low dimension space, the noninteracting far points are mapped as the local neighbour points[17], which destroys the results of low dimensional embedding. In addition, the LLE algorithm requires that a data set is sampled from a low-dimensional manifold in uniform density [18] and that it is vulnerable to noise. Thus, DWT was combined with LLE for feature extraction of MI-EEG, and the proposed method was called DWT-LLE. Considering the analysis above, the approximation of A_3 coefficients was chosen to reduce the dimensionality by LLE, and the nonlinear features of F_3 were defined as

$$F_3 = Y_3^* \in R^{3 \times 1} \quad (13)$$

A 3-D visualization of F_3 is shown in Fig. 4.

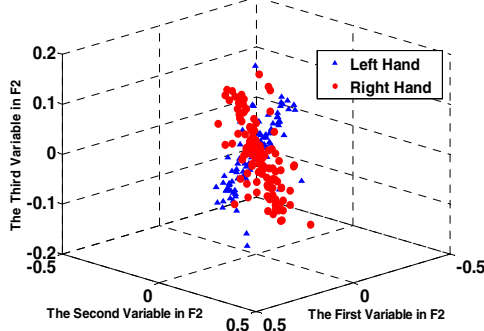


Fig. 3 3-D Visualization of Feature F_2

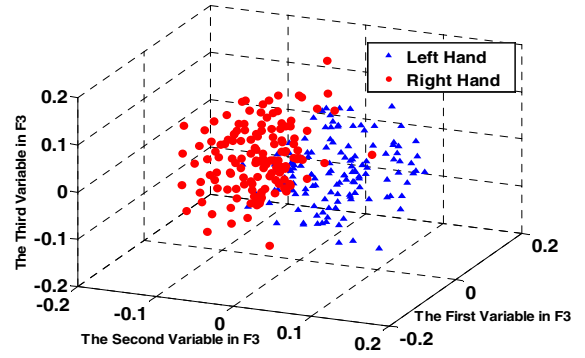


Fig. 4 3-D Visualization of Feature F_3

Fig. 3 shows that the low-dimensional embedding of F_2 presents a clustering distribution in the 3-D space, and the distribution of nonlinear features from two classes of signal cross each other. Further, we know from Fig 4 that compared with F_2 , the distribution of F_3 in the 3-D space shows better separability. We can thus conclude that there is an intimate connection between the nonlinear information contained in the MI-EEG and wavelet coefficients, which reflects the energy distribution of the signal in the time-frequency domain.

IV. PATTERN CLASSIFICATION

For objective evaluation of the quality of the extracted features, a Back-Propagation (BP) neural network was chosen as a classifier for the pattern recognition of the MI-EEG signal. Although the BP neural network has good self-learning abilities and strong adaptive capabilities, the random selection of its initial weights and thresholds results in its vulnerability to the local optimum problem, thus affecting the generalization ability of the classifier. So a three-layer structure BP neural network optimized by the Genetic Algorithm (GA) was used as a classifier and denoted by GA-BP.

A. The Structure of GA-BP

The number of input-layer neurons was decided by the feature dimension of the EEG, and the output-layer neuron was 1. The number of hidden-layer neurons was determined by the optimization of GA within a certain range, and the optimal value was 28. The transfer functions for the hidden-layer and output-layer were 'tansig' and 'pruelin', respectively, and the training function was 'trainlm'. Mean Square Error (MSE) was chosen as the performance function.

B. The Optimization Procedure of GA-BP

The optimization procedure for the weights and threshold for the input-layer and hidden-layer in the BP neural network was as follows:

Step 1: make a serial connection between the weight/bias of the input-layer and hidden-layer and the hidden-layer and output-layer of the candidate solution, called either individual, creature, or phenotype;

Step 2: transform it into an array of bits in binary encoding and set the array digits to 20;

Step 3: the population of candidate solutions is set to 40, and the maximum generation of evolution is 100.

Additionally, the probabilities of crossover and mutation are set to 0.7 and 0.05, respectively;

Step 4: use training samples and testing samples for network training and testing, respectively, and test that the error is regarded as an individual fitness value for each individual. The smaller the value is, the more outstanding the individual is.

Step 5: perform a heuristic search in the solution domain of genetic representation and use the fitness value of the individual to evaluate the solution by techniques inspired by natural evolution, such as inheritance, mutation, selection, and crossover. As a result, we can obtain a BP neural network optimized by GA with a better set of weight/bias.

V. EXPERIMENTAL RESULTS

A three-layer Back-Propagation(BP) neural network and the three-layer GA-BP networks were chosen as classifier for the pattern recognition of the MI-EEG signal, and the number of input-layer neurons was decided by the feature dimension of the MI-EEG. The number of output-layer neurons was 1. The number of hidden-layer neurons was determined by an optimization experiment within a certain range. To eliminate the contingency in the feature extraction process and to increase the objectivity of quality evaluation of features, 10-fold Cross Validation was used for estimating the performance of classification, which means that we split the data into 10 subsets, each of which were used as a validation set.

A. Feature Fusion

The features extracted by different methods can provide different information contained in MI-EEG, while there might be a problem of information redundancy and feature matching among different types of features. To obtain a feature vector containing comprehensive essential information, it is necessary that consideration should be given to the nonlinear nature and the ERD and ERS phenomena of MI-EEG. To find the optimal combinations between the time-frequency feature F_1 and nonlinear feature F_2, F_3 , various feature combinations are conducted using the GA-BP neural network as a classifier in a test with 10-fold cross validation. The average classification accuracy and variance are shown in Fig.5, where “+” represents feature fusion in serial order.

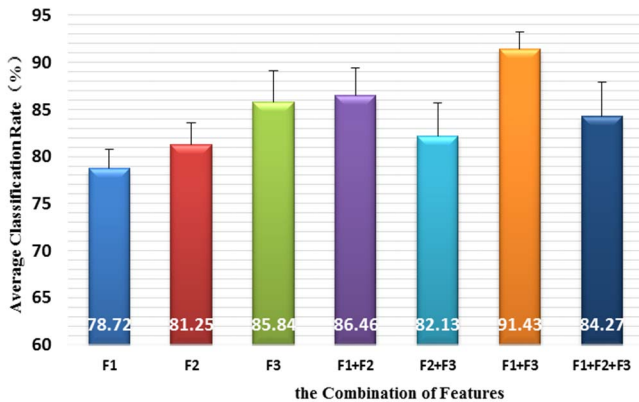


Fig. 5 Experimental results of the average classification accuracy and variation range in different feature combinations

Fig.5 illustrates that we can obtain the best average classification accuracy (91.43%) and the smallest standard deviation when feature F_1 is concatenated with feature F_3 , meaning that the two types of features show perfect complementarity and stability and the least redundant information.

B. Comparison of Multi Recognition Methods

For MI-EEG, a comparative study was performed with the DWT-based feature extraction methods, which have recently been proposed; the corresponding classifier; the feature dimension; and the optimal classification accuracy, as shown in Table.1.

Table 1 Comparison with Other DWT-based Feature Extraction Methods

number of reference	feature extraction	classifier	feature dimension	Top classification rate (%)
4	DWT-AR	LDA	24	92.14
6	DWT	LDA	20	91.00
19	DWT-ICA	BP	10	90.50
5	DWT-ICA	GA-SVM	14	90.71
8	DWT-SampEn	SVM	5	91.43
20	DWT-PSD	KNN	1	84.29
this paper	DWT-LLE	GA-BP	9	93.20

Note: “-” represents the combination or optimization of methods for feature extraction or classification.

Table.1 shows that the proposed method, called DWT-LLE, represents a best top-classification with fewer feature dimensions compared with recent ones.

C. Comparison of Multi DWT-base Feature Extraction Methods

To further verify the effectiveness of the proposed method for feature extraction and to evaluate the matching degree between the features and classifier, some comparative experiment studies were conducted based on the proposed method and traditional DWT-based methods were conducted on the same dataset. The classification performance was evaluated by the BP neural network and GA-BP neural network. The evaluation results of the experiments are listed in Fig.6, where “-” represents the combination of methods for feature extraction and the test was 10-fold cross validation.

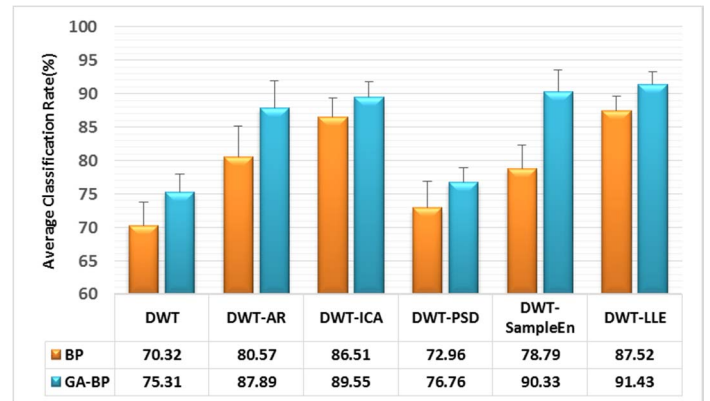


Fig. 6 Comparison of the average classification accuracy for multi DWT-based feature extraction methods

In the six experiments shown in Fig.6, the optimal numbers of hidden-layer neurons were 26, 30, 25, 7, 18 and

28. Fig. 6 shows that the method proposed in this paper had the best performance with a top average classification accuracy of 91.43% and a smaller variance of approximately $\pm 2\%$ around the average classification rate. The results demonstrate that the validity of this method for feature extraction of MI-EEG is encouraging and the proposed method demonstrates a high degree of matching with the GA-BP neural network.

VI. CONCLUSIONS

Regarding MI-EEG, the proposed feature extraction method based on LLE and DWT outperforms the existing methods in classification accuracy. The results of our experiments show that: (1) the time-frequency features based on DWT and the nonlinear features based on the LLE algorithm are highly complementary in information for classification and they contain less redundant data compared to features extracted by other methods; (2) the application of Manifold Learning as a nonlinear dimension reduction method enables us to find that high-dimensional MI-EEG signals show a significant clustering distribution in the low dimensional space; (3) the nonlinear information in the MI-EEG signals are mainly located in specific sub-bands related to certain movement imagine tasks, and the information can be represented by wavelet coefficients that depict the time-frequency energy distributions of the original signal. The nonlinear characteristics extracted by the method proposed in this paper can promote the further analysis of MI-EEG, paving the way for expanded applications of manifold learning to brain-computer research and posing a novel perspective of the nature of MI-EEG. However, the proposed method may not show its superiority for multi-class classification, so the organic combination with the method in [21] may provide an instructive solution to the problem.

ACKNOWLEDGMENT

The authors would like to thank the reviewers for their helpful comments and suggestions to improve this paper. We also thank researchers including G.E. Hinton, L. Maaten for their great achievements in dimension reduction and visualization, inspiring us to study related applications in BCI.

This work is financially supported by the National Natural Science Foundation of China (No.81471770), the Natural Science Foundation of Beijing (No.7132021) and Promotion Projects of Beijing University of Technology.

REFERENCES

- [1] J. Virts, "The third international meeting on brain-computer interface technology: making a difference," *IEEE Trans Neural Syst Rehabil Eng*, vol.14, no.2, pp.126-127, 2006.
- [2] B. Xu, "Feature extraction and classification of single trial motor imagery EEG", *Journal of Southeast University*, vol.37, no.4, pp.629-633, 2007.
- [3] N. Boonnak, S. Kamonsantiroj, "Wavelet Transform Enhancement for Drowsiness Classification in EEG Records Using Energy Coefficient Distribution and Neural Network," *International Journal of Machine Learning and Computing*, vol. 5, no. 4, pp. 288-293, August 2015.
- [4] B. Xu, "Feature Extraction and Classification of EEG in Online Brain-Computer Interface," *Acta Electronica Sinica*, vol. 39, no.5, pp.1025-1030, 2011.
- [5] Y. Wang, "Feature Extraction of Motor Imagery Electroencephalography Based on Time-frequency-space Domains," *Journal of Biomedical Engineering*, vol.31, no.5, pp.955-961, 2014.
- [6] M. Fira, "Classifications of Motor Imagery Tasks in Brain Computer Interface Using Linear Discriminant Analysis," *International Journal of Advanced Research in Artificial Intelligence*, vol.3, no.7, pp.5-9, 2014.
- [7] R. K. Chaurasiya, "Statistical Wavelet Features, PCA, and SVM Based Approach for EEG Signals Classification," *International Journal of Electrical, Computer, Electronics and Communication Engineering*, vol.9, no.2, pp.182-186, 2015.
- [8] Y. Zhang, "EEG feature extraction method based on wavelet transform and sample entropy," *CAAI Transactions on Intelligent Systems*, vol.7, no.4, pp.339-344, 2012.
- [9] X. Hao, "Correlation Analysis of Nonlinear Characteristics in EEG with CRS-R Score and Visual Characterization of Rehabilitation Process in DOC Patients," *Chinese Journal of Biomedical Engineering*, vol.34, no.2, pp.153-159, 2015.
- [10] F. Lee, R. Scherer, and R. Leeb, "Feature mapping using PCA, locally linear embedding and isometric feature mapping for EEG-based brain computer interface," *Digital Imaging in Media and Education*, pp.189-196, 2004.
- [11] Y. Pan, S. Ge, and A. Mamun, "Detection of seizures in EEG signal using weighted locally linear embedding and SVM classifier," *2008 IEEE Conference on Cybernetics and Intelligent Systems*, pp.358-363, 2008.
- [12] M. Mirsadeghi, H. Behnam, and R. Shalbaf, "Characterizing Awake and Anesthetized States Using a Dimensionality Reduction Method," *Journal of Medical Systems*, pp.13-21, JAN 2016.
- [13] X. Geng, D. Zhan, and Z. Zhou, "Supervised nonlinear dimensionality reduction for visualization and classification," *IEEE Transactions on Systems Man and Cybernetics Part B-Cybernetics*, vol.35, no.6, pp.1098-1107, 2005.
- [14] T. Sam, and K. Lawrence, "Nonlinear Dimensionality Reduction by Locally Linear Embedding," *Science*, pp.2323-2326, 2000.
- [15] H. Adeli, Z. Zhou, and N. Dadmehr, "Analysis of EEG records in an epileptic patient using wavelet transform," *Journal of Neuroscience Methods*, vol.123, pp.69-87, 2003.
- [16] G. Pfurtscheller, F. Lopes, "Event-related EEG/MEG synchronization and desynchronization: basic principles," *Clinical Neurophysiology*, vol.110, no.11, pp.1842-1857, 1999.
- [17] Y. Lei, "The Study of Manifold Learning Algorithms and Their Applications," Hefei: University of Science and Technology of China, 2011.
- [18] K. Lawrence, "Think Globally, Fit Locally: Unsupervised Learning of Low Dimensional Manifolds," *Journal of Machine Learning Research*, vol.4, pp.119-155, 2003.
- [19] P. Wang, "Feature Extraction of EEG for Imagery Left-Right Hands Movement," *Chinese Journal of Sensors and Actuators*, vol.23, no.9, pp.1220-1225, 2010.
- [20] P. Varshney, "Modelling of EEG Data for Right and Left Hand Movement," *International Journal of Engineering and Technical Research*, Special Issue, pp.193-197, 2014.
- [21] M. J. Khan, M. M. Hong, and K.-S. Hong, "Decoding of four movement directions using hybrid NIRS-EEG brain-computer interface," *Frontiers in Human Neuroscience*, vol. 8, article no. 244, pp. 1-10, 2014.